

Solutions to Tutorial 3

1.

$\vec{F}$ is conservative when there exists a non-singular potential function $V(x)$ satisfying

$$\vec{F} = -\vec{\nabla} V$$

(a)

$$F_x = -\frac{\partial V}{\partial x} = ayz + bx + c$$

$$\Rightarrow V = -ayzx - \frac{bx^2}{2} - cx + f_1(y, z) \rightarrow (i)$$

$$F_y = -\frac{\partial V}{\partial y} = axz + bz$$

$$\Rightarrow V = -axyz - byz + f_2(x, z) \rightarrow (ii)$$

$$F_z = -\frac{\partial V}{\partial z} = axy + by$$

$$\Rightarrow V = -axyz - byz + f_3(x, y) \rightarrow (iii)$$

From (i), (ii) and (iii) we find that:

$$V = -axyz - byz - cx - \frac{bx^2}{2} + A$$

where A is an arbitrary constant.

(b) Using the same method as in (a)

$$V = -z e^{-x} - y \ln z + C$$

(c) $\vec{F} = \frac{a \hat{r}}{r} = -\vec{\nabla} V$

Note:

$$\begin{aligned} \vec{\nabla} f(r) &= \frac{\partial f(r)}{\partial x} \hat{x} + \frac{\partial f(r)}{\partial y} \hat{y} + \frac{\partial f(r)}{\partial z} \hat{z} \\ &= \frac{\partial f(r)}{\partial r} \frac{\partial r}{\partial x} \hat{x} + \frac{\partial f(r)}{\partial r} \frac{\partial r}{\partial y} \hat{y} + \frac{\partial f(r)}{\partial r} \frac{\partial r}{\partial z} \hat{z} \\ &= \frac{\partial f(r)}{\partial r} \frac{x}{r} \hat{x} + \frac{\partial f(r)}{\partial r} \frac{y}{r} \hat{y} + \frac{\partial f(r)}{\partial r} \frac{z}{r} \hat{z} \\ &= \frac{1}{r} \frac{\partial f(r)}{\partial r} \vec{r} \\ &= \hat{r} \frac{\partial f(r)}{\partial r} \end{aligned}$$

Thus, $\vec{F} = \frac{a \hat{r}}{r} = -\vec{\nabla} V(r) = -\hat{r} \frac{\partial V}{\partial r}$

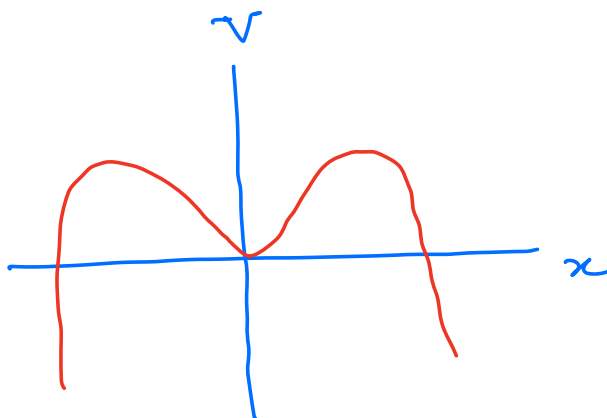
$$\Rightarrow \frac{\partial V}{\partial r} = -\frac{a}{r} \Rightarrow \boxed{V = -a \ln r}$$

All the force in (a), (b) and (c) are conservative.

2.

(a)
$$F(x) = - \frac{dV}{dx} = - \frac{4V_0 x}{a^2} \left(1 - \frac{x^2}{a^2} \right)$$

(b)



- When $F = 0$ there is equilibrium
- When $\frac{dF}{dx} < 0$, V has a local minimum and it is stable.
- When $\frac{dF}{dx} > 0$, V has a local maximum and it is unstable.

Here,

$x = a$ and $x = -a$
are unstable equilibrium positions.
 $x = 0$ is the stable equilibrium position.

(c)

Around the origin:

$$F \approx - \frac{4V_0 x}{a^2} = -kx$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{4V_0}{ma^2}}$$

(d) In order to escape to infinity from $x=0$, the particle needs to get at least to peak of the potential (refer to sketch above)

$$\frac{m v_{\min}^2}{2} = v_{\max} = v_0$$

$$\Rightarrow v_{\min} = \sqrt{\frac{2v_0}{m}} //$$

3.

(a) (i) attractive force

$$F_a = -Bx$$

$$v_a(x) - v_a(0) = - \int_0^x F_a dx' = Bx$$

$$\Rightarrow v_a(x) = Bx + v_a(0)$$

(ii) repulsive force

$$F_r = \frac{A}{x^2}$$

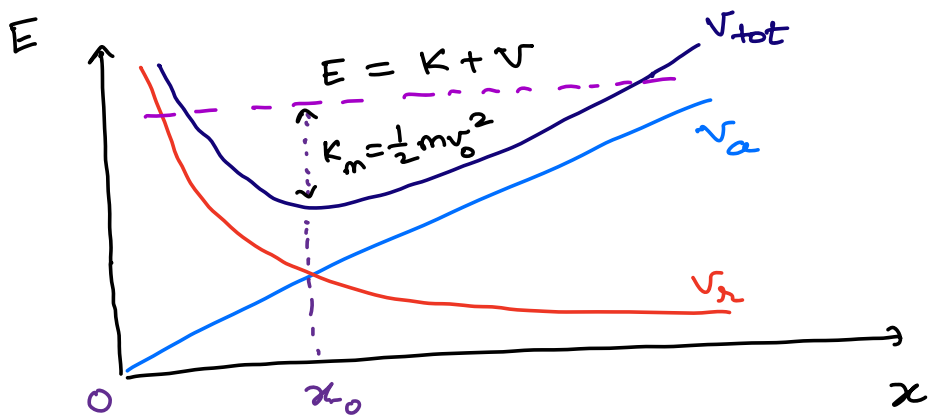
$$V_r(x) - V_r(\infty) = - \int_{\infty}^x A \frac{dx'}{x'^2} = \frac{A}{x}$$

$$\Rightarrow V_r(x) = \frac{A}{x}$$

Thus, $V_{\text{total}} = Bx + \frac{A}{x}$

(b)

$$B \leftarrow \textcircled{m} \rightarrow A/x^2$$



(c)

$$\frac{dV_{\text{total}}}{dx} = B - \frac{A}{x^2}$$

$$\left. \frac{dV_{\text{total}}}{dx} \right|_{x=x_0} = 0$$

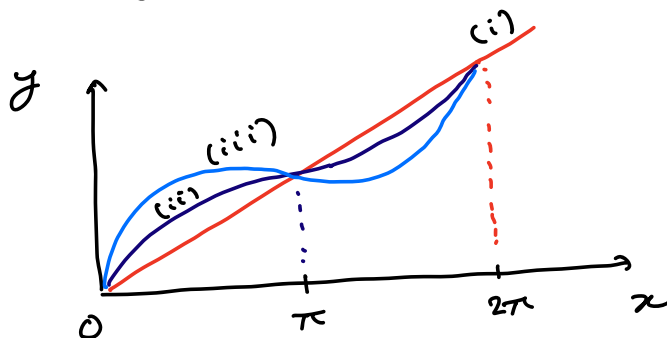
$$\Rightarrow x_0 = \sqrt{\frac{A}{B}}$$

4.

$$y(x) = x$$

Add $\eta(x) = \sin x$, to $y(x)$ to construct neighboring varied paths:

$$y(\alpha, x) = x + \alpha \sin x$$



Path (i) is for $\alpha = 0$

Path (ii) and (iii) are two different nonvanishing values of α .

$\eta(x) = \sin x$ obeys the end point conditions:
 $\eta(0) = 0 = \eta(2\pi)$

Now,
$$\frac{dy(\alpha, x)}{dx} = 1 + \alpha \cos x$$

Then,
$$F = \left(\frac{dy(\alpha, x)}{dx} \right)^2$$
$$= 1 + 2\alpha \cos x + \alpha^2 \cos^2 x$$

From,

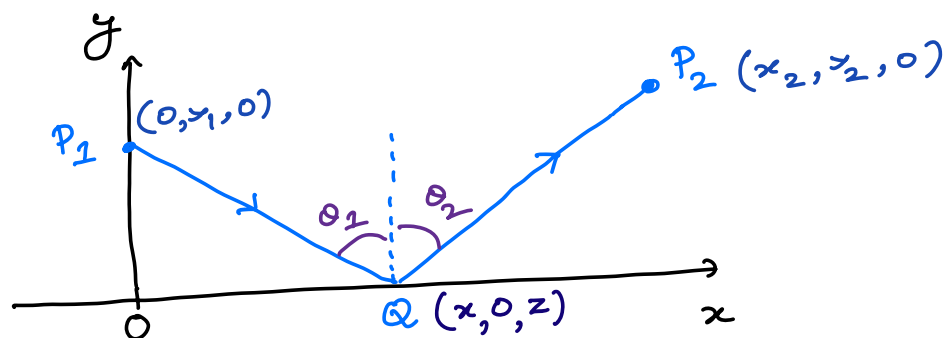
$$S(\alpha) = \int_{x_1}^{x_2} f\{y(\alpha, x); y'(\alpha, x); x\} dx$$

We get:

$$S(\alpha) = \int_0^{2\pi} (1 + 2\alpha \cos x + \alpha^2 \cos^2 x) dx$$
$$= 2\pi + \alpha^2 \pi$$

Clearly, $S(\alpha)$ is always greater than $S(0)$, no matter what value for α is chosen. $\Rightarrow$ Stationary value of $S(\alpha)$ occurs for $\alpha = 0$.

5.



We know that:

The actual path is a straight line within one medium.

$\Rightarrow$ segments from P1 to Q and from Q to P2 are straight.

Now, $P_1Q = \sqrt{x^2 + y_1^2 + z^2}$

$QP_2 = \sqrt{(x-x_2)^2 + y_2^2 + z^2}$

(Taking the mirror to be in the xz plane)

The total time, T, for the journey

P1QP2 is

$$T = \frac{\sqrt{x^2 + y_1^2 + z^2} + \sqrt{(x-x_2)^2 + y_2^2 + z^2}}{c}$$

To find the position of $Q = (x, 0, z)$ for which T is minimum we must have:

$$\frac{\partial T}{\partial z} = 0 \quad \text{and} \quad \frac{\partial T}{\partial x} = 0$$

Now,

$$\frac{\partial T}{\partial z} = 0 \Rightarrow \frac{z}{c \sqrt{x^2 + y_1^2 + z^2}} + \frac{z}{c \sqrt{(x-x_2)^2 + y_2^2 + z^2}} = 0$$

$$\Rightarrow z = 0$$

$\Rightarrow$ Q must lie in the same vertical plane as P_1 and P_2 .

$$\frac{\partial T}{\partial x} = 0 \Rightarrow \frac{x}{c \sqrt{x^2 + y_1^2 + z^2}} + \frac{x - x_2}{c \sqrt{(x-x_2)^2 + y_2^2 + z^2}} = 0$$

$$\Rightarrow \frac{x}{\sqrt{x^2 + y_1^2}} = \frac{x_2 - x}{\sqrt{(x-x_2)^2 + y_2^2}}$$

$$\Rightarrow \sin \theta_1 = \sin \theta_2$$

$$\Rightarrow \theta_1 = \theta_2$$

//

Solution to Practice problems

6.

$$\begin{aligned} v &= - \int F \, dz \\ &= \frac{2mMG}{a^2} \int \left(1 - \frac{z}{(a^2+z^2)^{1/2}} \right) dz \\ &= \frac{2mMG}{a^2} \left[z - (a^2+z^2)^{1/2} \right] \end{aligned}$$

Energy conservation:

$$\frac{1}{2}mv^2 + \frac{2mMG}{a^2} \left[z - (a^2+z^2)^{1/2} \right] = E, \text{ const.}$$

$$v = \dot{z}$$

$$v = 0 \quad \text{when} \quad z = \frac{4a}{3}$$

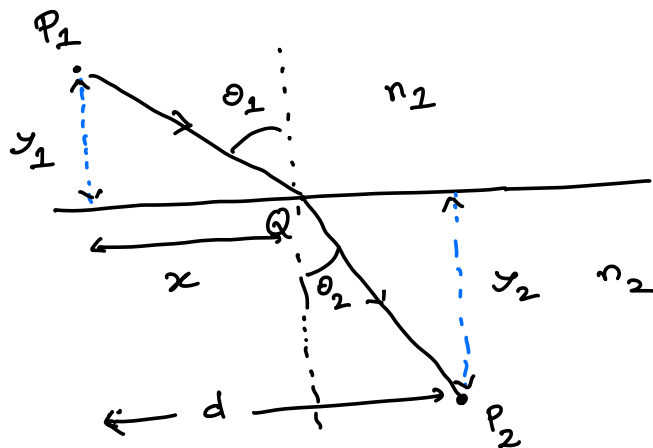
$$\begin{aligned} \text{So,} \quad E &= 0 + \frac{2mMG}{a^2} \left[\frac{4a}{3} - \frac{5a}{3} \right] \\ &= - \frac{2mMG}{3a} \end{aligned}$$

$$\text{Thus,} \quad \frac{1}{2}mv^2 = \frac{2mMG}{a^2} \left[(a^2+z^2)^{1/2} - z \right] - \frac{2mMG}{3a}$$

Putting $z=0$ into the above expression we find that the speed of the particle when it hits the disk is

$$v|_{z=0} = \left(\frac{8MG}{3a} \right)^{1/2} //$$

7.



The total time taken from the point P_1 to the point P_2 is

$$T = \frac{\sqrt{x^2 + y_1^2}}{c/n_1} + \frac{\sqrt{(d-x)^2 + y_2^2}}{c/n_2}$$

$$= \frac{1}{c} \left[n_1 \sqrt{x^2 + y_1^2} + n_2 \sqrt{(d-x)^2 + y_2^2} \right]$$

Fermat's principle requires that:

$$\frac{dT}{dx} = 0$$

$$\Rightarrow \frac{n_1 x}{\sqrt{x^2 + y_1^2}} - \frac{n_2 (d-x)}{\sqrt{(d-x)^2 + y_2^2}} = 0$$

$$\Rightarrow n_1 \sin \theta_1 = n_2 \sin \theta_2 //$$